THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · HARD LEVEL MCQ TEST SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. From the top of a tower $6\sqrt{3}$ m high, the angles of depression of two objects on level ground, on the same side of the tower and in line with its foot, are 30° and 60°. The distance between the two objects is: 1

(a) 24 m **(b)** $12\sqrt{3}$ m **(c)** 6 m **(d)** 12 m

2. Two concentric circles have radii 68 cm and 32 cm. A chord of the larger circle touches the smaller circle. The length of the chord is: 1

(a) 60 cm **(b)** 120 cm **(c)** 72 cm **(d)** 100 cm

3. A sector of a circle of radius 42 cm has central angle 45°. The perimeter of the sector is: 1

(a) 75 cm **(b)** 117 cm **(c)** 159 cm **(d)** 33 cm

4. A solid cone and a solid hemisphere have equal base radii 7 cm and equal volumes. The height of the cone is: 1

(a) 10.5 cm **(b)** 14 cm **(c)** 21 cm **(d)** 7 cm

5. For the data with classes 20–40, 40–60, 60–80, 80–100, 100–120, 120–140, 140–160 with frequencies 3, 13, 11, 17, 15, 24, 17 respectively, The median of the data is: 1

(a) 120 **(b)** 108 **(c)** 100 **(d)** 103.6

6. A bag contains 7 red and 9 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{4}{7}$. The value of x is: 1

(a) 4 **(b)** 10 **(c)** 5 **(d)** 6

7. Which of these is a rational number? 1

(a) $\sqrt{2}+\sqrt{3}$ **(b)** $2\sqrt{5}$ **(c)** $\dfrac{\sqrt{12}}{\sqrt{3}}$ **(d)** $\sqrt{4}$

8. For the cubic with zeroes $1, 2, 3$, $\alpha\beta+\beta\gamma+\gamma\alpha$ is: 1

(a) 36 **(b)** 11 **(c)** -11 **(d)** 6

9. If $x+y=7$ and $xy=10$, then $x^2+y^2=$ 1

(a) 29 **(b)** 49 **(c)** 39 **(d)** 69

10. The equation $kx^2+6x+3=0$ has real roots when: 1

(a) $k \geq 3$ **(b)** $k<3$ only

(c) $k \leq 3$ and $k \neq 0$ **(d)** $k>0$

11. The sum of the series $5 + 9 + 13 + \cdots + 101$ is: 1

(a) 1325 **(b)** 1430 **(c)** 2650 **(d)** 1224

12. In $\triangle ABC$ right-angled at B, $AB = 3$, $BC = 4$. If $BD \perp AC$, then AD (in cm) is: 1

(a) $\frac{16}{5}$ **(b)** $\frac{3}{2}$ **(c)** $\frac{9}{5}$ **(d)** $\frac{12}{5}$

13. The side of the square with diagonal endpoints $(0, 0)$ and $(6, 8)$ has squared length: 1

(a) 25 **(b)** 200 **(c)** 100 **(d)** 50

14. $\dfrac{\sin^2 25^\circ + \sin^2 65^\circ}{\cos^2 35^\circ + \cos^2 55^\circ} =$ 1

(a) $\frac{1}{2}$ **(b)** 2 **(c)** 0 **(d)** 1

15. From the top of a building 3 m high, the angle of elevation of the top of a tower is 30° and the angle of depression of its foot is 60°. The height of the tower is: 1

(a) 4 m **(b)** $4\sqrt{3}$ m **(c)** $\sqrt{3}$ m **(d)** 1 m

16. In $\triangle ABC$, right-angled at B, $AB = 45$ cm and $BC = 60$ cm. A circle is inscribed in the triangle. The radius of the inscribed circle is: 1

(a) 30 cm **(b)** 15 cm **(c)** 16 cm **(d)** 37 cm

17. An arc of length 22 cm subtends an angle at the centre of a circle of radius 42 cm. The angle subtended by the arc at the centre is: 1

(a) 15° **(b)** 60° **(c)** 30° **(d)** 150°

18. The radii of two solid spheres are in the ratio 2 : 3. The ratio of their surface areas is: 1

(a) 2 : 3 **(b)** 8 : 27 **(c)** 4 : 9 **(d)** 8 : 9

19. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100, 100–110 with frequencies 17, 21, 24, 17, 6, 4, 11 respectively, The mode of the data is: 1

(a) 68 **(b)** 60 **(c)** 63 **(d)** 70

20. A bag contains 11 red and 5 blue balls. 9 more red balls are put into the bag and a ball is drawn at random. The probability that the ball drawn is red is: 1

(a) $\frac{21}{25}$ **(b)** $\frac{1}{5}$ **(c)** $\frac{4}{5}$ **(d)** $\frac{19}{25}$

21. A number ends with exactly 2 zeros if its prime factorisation (other factor 3) is: 1

(a) $2^3 \times 5^2 \times 3$ **(b)** $2^3 \times 5^1 \times 3$ **(c)** $3^2 \times 7$ **(d)** $2^1 \times 5^2 \times 3$

22. If the zeroes of $x^2 - 10x + k$ differ by 4, then k is: 1

(a) 29 **(b)** 21 **(c)** 84 **(d)** 42

23. If $5x + 3y = 47$ and $3x + 5y = 41$, then $x - y =$ 1

(a) 6 **(b)** 44 **(c)** 3 **(d)** $\frac{3}{4}$

24. If α, β are roots of $x^2 - 5x + 6 = 0$, then $\dfrac{1}{\alpha} + \dfrac{1}{\beta} =$ 1

(a) 30 **(b)** $\frac{5}{6}$ **(c)** $-\frac{5}{6}$ **(d)** $\frac{6}{5}$

25. How many terms of the AP $20, 16, 12, 8, \ldots$ give sum 0? 1

(a) 12 **(b)** 11 **(c)** 22 **(d)** 10

26. A square has diagonal $5\sqrt{2}$ cm. Its area (cm²) is: 1

(a) 25 **(b)** 50 **(c)** 10 **(d)** 100

27. The three points $(0,0)$, $(8,0)$ and $(3,7)$ enclose area: 1

(a) 56 **(b)** 14 **(c)** 42 **(d)** 28

28. The value of $\tan 10^\circ \tan 20^\circ \tan 70^\circ \tan 80^\circ$ is: 1

(a) 1 **(b)** 2 **(c)** $\tan 45^\circ + 1$ **(d)** 0

29. Two points on level ground lie on opposite sides of a tower of height 9 m, in line with its foot. The angles of elevation of the top of the tower from the two points are 45° and 60°. The distance between the two points is: 1

(a) 12 m **(b)** $9 - 3\sqrt{3}$ m **(c)** $9 + 9\sqrt{3}$ m **(d)** $9 + 3\sqrt{3}$ m

30. A point P is 85 cm from the centre O of a circle of radius 77 cm. The length of the tangent drawn from P to the circle is: 1

(a) 36 cm **(b)** 8 cm **(c)** 1296 cm **(d)** 162 cm

31. A square is inscribed in a circle of radius 35 cm. The area of the circle outside the square is: 1

(a) 2625 cm² **(b)** 3850 cm² **(c)** 2450 cm² **(d)** 1400 cm²

32. The radii of two solid spheres are in the ratio 2 : 3. The ratio of their volumes is: 1

(a) 8 : 27 **(b)** 4 : 9 **(c)** 16 : 27 **(d)** 2 : 3

33. For a distribution with median class 20–30, $N = 50$, $cf = 18$ and $f = 14$, the median is: 1

(a) 22.5 **(b)** 28 **(c)** 25 **(d)** 24

34. A box contains 70 tickets numbered 1 to 70. One ticket is drawn at random. The probability that the number drawn is a prime number is: 1

(a) $\frac{51}{70}$ **(b)** $\frac{9}{35}$ **(c)** $\frac{2}{7}$ **(d)** $\frac{19}{70}$

35. The LCM of 36, 48 and 72 is: 1

(a) 144 **(b)** 1728 **(c)** 5184 **(d)** 72

36. If $x = 2$ is a zero of $x^2 - 9x + 14$, the other zero is: 1

(a) -7 **(b)** 7 **(c)** 11 **(d)** 12

37. The pair $kx + 3y = k - 3$ and $12x + ky = k$ has no solution when $k =$ 1

(a) 3 **(b)** -6 **(c)** 6 **(d)** 12

38. If the roots of $x^2 - 9x + 14 = 0$ are α, β, then $\alpha^2 + \beta^2 =$ 1

(a) 53 **(b)** 109 **(c)** 81 **(d)** 67

39. If the 5th term of an AP is 11 and the 11th term is 5, then the 16th term is: 1

(a) 16 **(b)** 0 **(c)** 1 **(d)** 55

40. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 2 : 3$. The ratio of areas of $\triangle ADE$ to trapezium $DBCE$ is 4: 1

(a) 29 **(b)** 21 **(c)** 9 **(d)** 25

41. The point on the y-axis equidistant from $(1,2)$ and $(5,4)$ has $y =$ 1

(a) -9 **(b)** 9 **(c)** 3 **(d)** 18

42. If $\sin\theta = \frac{12}{13}$ then $\sin^2\theta + \cos^2\theta + \tan^2\theta =$ 1

(a) $\frac{25}{169}$ **(b)** $\frac{169}{25}$ **(c)** $\frac{144}{169}$ **(d)** $\frac{144}{25}$

43. A person looks at the top of a vertical pole from a point on the ground. The top of the pole is $4\sqrt{3}$ m above the level of the person's eyes and the eyes are 12 m horizontally from the pole. The angle of elevation of the top of the pole is: 1

(a) 45° **(b)** 30° **(c)** $30\sqrt{3}^\circ$ **(d)** 60°

44. AB is a chord of a circle with centre O and PA, PB are tangents. Then the line OP is: 1

(a) equal to AB **(b)** a tangent to the circle

(c) the perpendicular bisector of AB **(d)** parallel to AB

45. A horse is tied at a corner of a large square field. The rope is first 7 m long and then lengthened to 35 m. The increase in the area it can graze is: 1

(a) 3696 m² **(b)** 616 m² **(c)** 924 m² **(d)** 962.5 m²

46. A solid cylinder of base radius 6 cm and height 10 cm is melted and recast into identical solid cones, each of base radius 3 cm and height 4 cm. The number of cones obtained is: 1

(a) 10 **(b)** 30 **(c)** 34 **(d)** 60

47. The mean of a distribution with classes 100–120, 120–140, 140–160, 160–180, 180–200, 200–220, 220–240, 240–260 and frequencies 19, 3, 4, 12, p, 13, 22, 7 respectively is 184.6. The value of p is: 1

(a) 20 **(b)** 22 **(c)** 19 **(d)** 25

48. A lot of 250 ball pens contains 35 defective pens. One pen is drawn at random from the lot. The probability that the pen is good (not defective) is: 1

(a) $\frac{108}{125}$ **(b)** $\frac{107}{125}$ **(c)** $\frac{43}{50}$ **(d)** $\frac{7}{50}$

49. The least number that is divisible by every number from 1 to 7 is: 1

(a) 2520 **(b)** 420 **(c)** 210 **(d)** 840

50. If α, β are zeroes of $x^2 - 6x + 8$, then $\alpha^3 + \beta^3$ is: 1

(a) 360 **(b)** 72 **(c)** 120 **(d)** 216

END OF QUESTION PAPER **Total: 50 Marks**

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